

Name _____ Key _____

Total 12 points

Please show all your work.

1. (2 pts) Determine whether the numerical value is a parameter or a statistic:

a) The average salary of 25 employees at a large company is \$33,000. Statistic

b) The average late fee for all CITI CARD credit card holders was found to be \$56.79.

Parameter

2. (4 pts) Determine whether the following data is qualitative or quantitative:

a) the numbers on the shirts of a girl's soccer team

qualitative

b) the number of patients that enter the clinic

quantitative

c) number of milligrams of tar in 30 cigarettes

quantitative

d) social security numbers of 15 students

qualitative

3. (4 pts) Identify the sampling technique used (random, cluster, stratified, convenience, systematic):

a) Every sixth person boarding a plane is searched thoroughly.

Systematic

b) At a local community college, five students from each of the ten sections of biology 1010 are interviewed.

Stratified

c) A researcher for an airline interviews all of the passengers on five randomly selected flights.

Cluster

d) To avoid working late, the quality control manager inspects the first 15 items produced that day.

Convenience

4. (2 pts) Decide which method of data collection (observational study, experiment, simulation, or survey) you are using to gather data for the following study and explain your reasoning:

You collect data on the ages of students at the U of U by interviewing a random sample of students on campus.

survey